

TS200 RTD Temperature Sensor Assembly

SKU / Model Number: TS200-PT100-3W

Manufacturer: ThermoPoint GmbH

Category: Sensors | ETIM Code: EC002988

PRODUCT OVERVIEW

The TS200 is a Class A Pt100 RTD temperature sensor assembly featuring a spring-loaded element for fast response and excellent vibration resistance. Designed for petrochemical and food-grade process applications.

TECHNICAL SPECIFICATIONS

Sensor Type	Pt100, 3-wire
Accuracy Class	Class A (IEC 60751)
Measurement Range	-200 to 600 deg C
Response Time (t63)	< 3 seconds in water
Self-heating Error	< 0.05 deg C
Insulation Resistance	> 100 MOhm at 500 V DC
Drift	< 0.05 deg C / 1000 hours

ELECTRICAL RATINGS

Excitation Current	1 mA recommended
Lead Resistance	< 2 ohm per lead
Connection Type	3-wire RTD
Cable Length	2 m standard
Shielding	Braided copper shield

DIMENSIONS & MECHANICAL

Probe Diameter	6 mm
Insertion Length	100 to 1000 mm
Connection Head	Form B (DIN 43729)
Process Connection	G 1/2 A (DIN 16288)
Weight	350 g (100 mm insertion)

OPERATING CONDITIONS

Ambient Temperature	-40 to 85 deg C
Process Temperature	-200 to 600 deg C
Max Pressure	100 bar at 20 deg C
Vibration	10 g (10-500 Hz)
IP Rating	IP68

MATERIALS OF CONSTRUCTION

Thermowell	316L Stainless Steel
Element Housing	Inconel 600
Seals	Graphite
Connection Head	Aluminum alloy, epoxy coated
Insulation	MgO (Magnesium Oxide)

PERFORMANCE DATA / SELECTION TABLE

Range (deg C)	Tolerance Class A	Tolerance Class B	Response t63
-200 to -100	+/- 0.35 deg C	+/- 0.70 deg C	< 5 s
-100 to 0	+/- 0.25 deg C	+/- 0.50 deg C	< 4 s
0 to 200	+/- 0.15 + 0.002*T	+/- 0.30 + 0.005*T	< 3 s
200 to 600	+/- 0.15 + 0.002*T	+/- 0.30 + 0.005*T	< 3 s

CERTIFICATIONS & APPROVALS

CE | ATEX II 1G | IECEx | FDA 21 CFR | RoHS

Note: Specific certification documents and test reports are available upon request. All products are manufactured under ISO 9001:2015 quality management system.

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